Pranav Senthilkumar

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Education

Texas A&M University College Station, TX
Bachelor of Science in Electrical Engineering, Engineering Honors *Expected May 2028*
- **Coursework:** Digital Logic, Machine Learning, Multivariable Calculus, Differential Equations

Technical Skills

Languages: SystemVerilog, Verilog, C++, Python, Java
EDA & Platforms: Vivado, FPGA, ESP32, Raspberry Pi, Arduino
Software & ML: Linux, Git, GitHub Actions, ROS 2, PyTorch, TensorFlow, ONNX

Experience

Consultancy Project Team Member – Shallow Sports Boats September 2026 – Present
Aggies Create *College Station, TX*
- Implementing C++ firmware on an ESP32 to stream GPS, IMU, sonar, and propulsion telemetry to the navigation stack
- Deploying a quantized vision model on a Raspberry Pi for real-time water segmentation and obstacle detection
- Designing C++ and ROS 2 navigation software that turns mission-level objectives into autonomous survey behavior

Robotics Software Engineering Intern August 2026 – September 2026
Nietzsche Labs *Remote*
- Shipped a cross-platform Electron app that provisions robot OS images onto SD cards, backed by an AWS serverless API
- Enforced 8 pre-write safety checks with target-drive re-verification to prevent accidental overwrites of host disks
- Automated first-boot partition expansion on embedded Linux images, reclaiming **90–97%** of storage stranded behind a fixed 2.75 GiB partition, validated on 10 card sizes (2–256 GB) with zero failures

Undergraduate Machine Learning Researcher June 2026 – Present
LIVE Lab, Texas A&M University *College Station, TX*
- Building ML models to turn regional wind forecasts into street-level predictions across urban and open terrain
- Developed an end-to-end urban wind pipeline combining LiDAR, terrain, and building data with a 3D model, cutting near-ground error by **50%** and prediction time from ∼3 min to under 1 s
- Improved 4× wind-field resolution by **59.9%** over interpolation using a residual correction model

Projects

RV32I RISC-V Processor | *SystemVerilog, Vivado, RISC-V* September 2026 – Present
- Developing a 32-bit RV32I RISC-V processor in SystemVerilog targeting the Zybo Z7-10 FPGA
- Designing a 5-stage pipelined datapath with an ALU, register file, control logic, and memory interface
- Building simulation and synthesis workflows in Vivado to verify functionality and measure timing and FPGA resource usage

POLITE AI Inference Accelerator August 2026 – Present
- Architecting a 4×4 INT8 systolic array sized via an area tradeoff study against a ~ 2 mm^2 SKY130 budget
- Designing weight-stationary PE dataflow with 32-bit accumulators and double-buffered memory to hide 4-cycle weight-load latency behind compute
- Writing cycle-accurate timing contracts (5-cycle fill, 5-cycle flush, 4-cycle drain) driving RTL and verification

Regdrift | *Python, CMSIS-SVD, pytest, GitHub Actions* June 2026 – July 2026
- Created a CI tool that blocks **22 classes** of breaking register-map changes before they surface as firmware faults
- Engineered deterministic comparisons of registers, bit fields, interrupts, and peripherals for reproducible reports
- Validated with **330 automated tests** across 15 microcontroller definitions from 5 semiconductor vendors

RISC-V RV32I ALU | *Verilog, RISC-V, Python, RTL Verification* June 2026
- Designed a 32-bit RV32I ALU in Verilog supporting 10 arithmetic, logic, shift, and comparison operations
- Implemented branch comparison hardware for 6 signed and unsigned RISC-V branch conditions
- Verified RTL with **27 directed cases** and a Python reference model generating reproducible randomized test vectors

Digital Logic Gate Simulator | *C++* December 2025
- Built a C++ digital logic simulator supporting 6 elementary gates: AND, OR, NOT, NAND, NOR, and XOR
- Modeled per-gate propagation delays to reproduce timing-dependent transitions across simulated logic circuits
- Generated waveform visualizations for input and output signals, exposing gate states and propagation timing over time

Leadership

Treasurer August 2026 – Present
Rev Silicon *College Station, TX*
- Managing budget and funding for a student silicon design organization pursuing its first chip fabrication on SKY130
- Coordinating hardware procurement and EDA software needs so members can take designs from concept to fabrication

Publication

Pranav Senthilkumar, et al. "Fine-Tuning Language Models for Ethical Ambiguity: A Comparative Study of Alignment with Human Responses." NeurIPS 2024 SoLaR Workshop | arXiv:2410.07826